

0.1 Question 1: Human Context and Ethics

In this part of the project, we will explore the human context of our housing dataset. **You should watch [Lecture 13](#) before attempting this question.**

0.1.1 Question 1a

Consider the following question: *“How much is a house worth?”*

Who might be interested in an answer to this question? Be sure to list at least three different parties (people or organizations) and state whether each one has an interest in seeing a low or high housing price.

Your response should be approximately 3 to 6 sentences.

1. Homeowners may prefer low assessed prices (lower taxes) but may prefer high prices when selling/refinancing.
2. County/municipal governments would prefer higher assessed house prices as this would mean more tax revenues.
3. Prospective buyers would prefer lower housing prices as this would allow them to save money and take out less mortgages.

0.1.2 Question 1b

Which of the following scenarios strikes you as unfair, and why? You can choose more than one. There is no single right answer, but you must explain your reasoning. Would you consider some of these scenarios more (or less) fair than others? Why?

- A. A homeowner whose home is assessed at a higher price than it would sell for.
- B. A homeowner whose home is assessed at a lower price than it would sell for.
- C. An assessment process that systematically overvalues inexpensive properties and undervalues expensive properties.
- D. An assessment process that systematically undervalues inexpensive properties and overvalues expensive properties.

Your response for each chosen scenario should be approximately 2 to 3 sentences.

C seems the most unfair to me because it is regressive—systematic over-taxation of lower-priced owners and breaks horizontal/vertical equity. This primarily affects the low-income family, which means they have to pay more than necessary for the property. A is also unfair because it inflates tax burden and can be arbitrary for those least able to appeal. However, compared to C, it might be less concerning as for C, the money of the low income families may be very crucial to the daily survival and maintainance. In the context of A, when the richer homeowners pay more tax than necessary, it may affect them not as severely as low-income families due to the larger amount of assets/money they possess. They are able to bear more risk as they can undertake relatively minor market fluctuation prices on their property. B/D have issues too, but C is the clearest systemic harm to me.

0.1.3 Question 1d

What were the central problems with the earlier property tax system in Cook County as reported by the Chicago Tribune? What were the primary causes of these problems?

Your response should be approximately 2 to 4 sentences.

Note: Along with reading the paragraph above, you will need to watch [Lecture 13](#) to answer this question.

Cook County's assessment system was both inaccurate and regressive. Expensive homes were often undervalued, while cheaper homes—common in Black and Latino neighborhoods—were overvalued. That meant working-class, nonwhite homeowners paid more than their fair share. Two big drivers: (1) opaque, poorly validated models that didn't correct these patterns, and (2) an appeals system that favored people with money and expertise. Wealthier owners could hire help to lower their assessments; people without those resources usually couldn't.

0.1.4 Question 1e

In addition to being regressive, how did the property tax system in Cook County place a disproportionate tax burden on non-white property owners?

Your response should be approximately 3 to 4 sentences.

Because the county needs to collect a fixed amount of property tax, when wealthier homeowners win appeals and lower their assessments, the tax burden shifts to everyone else. Combine that with a model that already overvalues lower-priced homes, and you get higher effective tax rates for working-class, nonwhite neighborhoods. On top of that, the appeals process itself is harder to access—less time, less help, and sometimes less physical access—so these communities are less likely to get relief, locking in the unfairness year after year.

0.2 Question 4a

We can assess a model's performance and quality of fit with a plot of the residuals ($y - \hat{y}$) versus the observed outcomes (y).

In the cell below, use `plt.scatter` ([documentation](#)) to plot the **model 2** residuals of Log Sale Price versus the original Log Sale Price values. For this part, you only need to plot the residuals and outcomes for the **validation data**.

- You should also **ensure that the dot size and opacity in the scatter plot are set appropriately** to reduce the impact of overplotting as much as possible. However, with such a large dataset, it is difficult to avoid overplotting entirely.

```
In [33]: residuals_m2 = Y_valid_m2 - Y_predicted_m2
plt.figure(figsize=(8,5))
plt.scatter(Y_valid_m2, residuals_m2, s=5, alpha=0.2)
plt.axhline(0, color='k', linewidth=1, linestyle='--')
plt.xlabel('Log Sale Price (Original)')
plt.ylabel('Residuals')
plt.title('Residuals vs Log Sale Price')
plt.show()
```



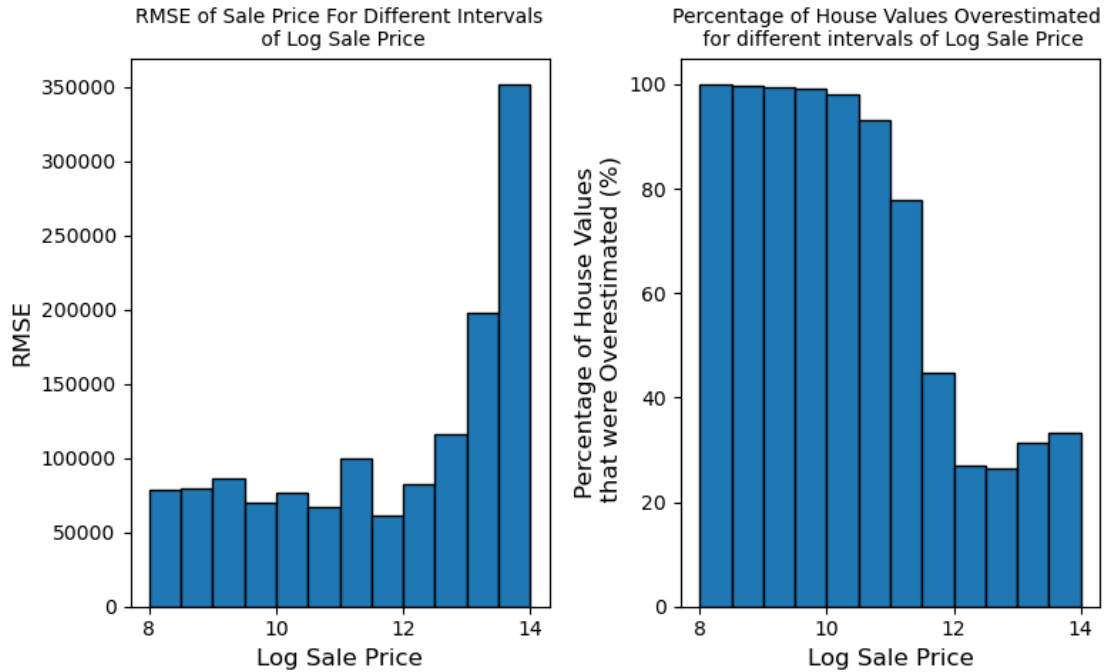
0.2.1 Question 6c

Using the functions above, we can generate visualizations of how the RMSE of sale price and proportion of overestimated houses vary for different intervals:

```
In [56]: # RMSE plot
plt.figure(figsize = (8,5))
plt.subplot(1, 2, 1)
rmse = []
for i in np.arange(8, 14, 0.5):
    rmse.append(rmse_interval(preds_df, i, i + 0.5))
plt.bar(x = np.arange(8.25, 14.25, 0.5), height = rmse, edgecolor = 'black', width = 0.5)
plt.title('RMSE of Sale Price For Different Intervals\n of Log Sale Price', fontsize = 10)
plt.xlabel('Log Sale Price')
plt.yticks(fontsize = 10)
plt.xticks(fontsize = 10)
plt.ylabel('RMSE')

# Overestimation plot
plt.subplot(1, 2, 2)
props = []
for i in np.arange(8, 14, 0.5):
    props.append(prop_overest_interval(preds_df, i, i + 0.5) * 100)
plt.bar(x = np.arange(8.25, 14.25, 0.5), height = props, edgecolor = 'black', width = 0.5)
plt.title('Percentage of House Values Overestimated \n for different intervals of Log Sale Price',
          fontsize = 10)
plt.xlabel('Log Sale Price')
plt.yticks(fontsize = 10)
plt.xticks(fontsize = 10)
plt.ylabel('Percentage of House Values\n that were Overestimated (%)')

plt.tight_layout()
plt.show()
```



Which of the two plots above would be more useful in ascertaining whether the assessments tended to result in progressive or regressive taxation? Provide a brief explanation to support your choice of plot.

Then, explain whether your chosen plot aligns more closely with scenario C or scenario D from q1b:

- C. An assessment process that systematically overvalues inexpensive properties and undervalues expensive
- D. An assessment process that systematically undervalues inexpensive properties and overvalues expensive

Your response should be approximately 3 to 4 sentences.

I'd use the overestimation percentage plot on the right. It directly shows whether each price bin is mostly over or under assessed (50% being the neutral line), which is what we need to judge progressive vs regressive behavior. In our plot, cheaper bins are > 50% overestimated while expensive bins are <50%, so the model overvalues inexpensive homes and undervalues expensive ones. That matches Scenario C and implies a regressive pattern.

0.3 Question 7: Evaluating the Model in Context

0.4 Question 7a

When evaluating your model, we used RMSE. In the context of estimating the value of houses, what does the residual mean for an individual homeowner? How does a positive or negative residual affect them in terms of property taxes? Discuss the cases where the residual is positive and negative separately.

Your response should be approximately 2 to 4 sentences.

A residual is $\text{Actual} - \text{Predicted}$ (in our case, sale price on the log scale). A positive residual means the model undervalued the home (assessed too low), which implies that the owner likely pays less tax than they should. A negative residual means the model overvalued the home (assessed too high), which follows that the owner likely pays more tax than they should. Looking at the pattern of residuals across price bins/areas tells us whether the system is progressive or regressive.

0.5 Question 7b

Reflecting back on your exploration in Questions 6 and 7a, in your own words, what makes a model's predictions of property values for tax assessment purposes "fair"?

This question is open-ended and part of your answer may depend on your specific model; we are looking for thoughtfulness and engagement with the material, not correctness.

Hint: Some guiding questions to reflect on as you answer the question above: What is the relationship between RMSE, accuracy, and fairness as you have defined it? Is a model with a low RMSE necessarily accurate? Is a model with a low RMSE necessarily "fair"? Is there any difference between your answers to the previous two questions? And if so, why?

Your response should be approximate 1 to 2 paragraphs. Feel free to answer the questions in the hint to structure your answer.

Low RMSE implies better average accuracy, but accuracy does not imply fairness. A model can have great RMSE overall and still be unfair if its errors aren't balanced. For example, consistently overvaluing cheaper homes and undervaluing expensive homes. That pattern raises effective tax rates for lower-priced (often non-white, working-class) neighborhoods even though the model "performs well." Henceforth, a "fair" tax model should aim for both a competitive RMSE and residual parity across price tiers and communities (residuals centered near 0, similar dispersion, ~50/50 over/under by group). In practice I'd monitor RMSE and distributional diagnostics (percent overestimated by bin, group-wise residual means/medians), and adjust features/procedures if we see systematic over-assessment of vulnerable groups.

0.6 Question 8: Finding Better Metrics For Our Model

0.7 Question 8a

As discussed in Project A1, RMSE—while a widely used and powerful error metric introduced in earlier lectures—is not always the most appropriate choice when evaluating the *fairness* of a property appraisal system. In Question 7, you already encountered some of RMSE’s limitations, particularly its tendency to disproportionately emphasize errors in high-priced properties due to the squaring of residuals.

In this question, rather than relying on RMSE, we will train and evaluate our model using another custom fairness metric. Specifically, we will examine the **Mean Absolute Percentage Error (MAPE)**, which measures the average error as a percentage of the true value. This allows us to better assess whether the model is making relatively fair predictions across different segments of the housing market.

In the code cells that follow, we’ll explore how MAPE varies across price ranges—comparing the model’s relative performance on inexpensive versus expensive housing. This helps identify whether the model systematically favors or disadvantages certain groups of properties based on their price.

Note: You’ll notice that we’re no longer using `lm.LinearRegression()`, but instead have started using `minimize`. This is because `scikit-learn` does not allow customization of the loss function in their standard linear regression model. The approach shown below provides an equivalent way to train a linear regression model using a custom error metric.

Warning: These cells take quite a long time to run. Please be patient, wait, and avoid restarting the kernel or rerunning these cells more than necessary.

```
In [57]: data = pd.read_csv("cook_county_train.csv", index_col='Unnamed: 0')
        trainX, trainY = feature_engine_final(data)
```

```
In [58]: # X is the design matrix (including bias column), y is the vector of true outputs, theta is th
        def mape(theta, X, y):
            y_pred = X @ theta # compute predicted values using linear combination of features
            percentage_error = np.abs((y - y_pred) / y) # calculate element-wise percentage errors
            return np.mean(percentage_error) # return the MAPE
```

```
In [59]: from scipy.optimize import minimize

        # Add bias (intercept) column to the design matrix
        trainX_with_bias = np.column_stack([np.ones(trainX.shape[0]), trainX])

        # Initialize parameter vector with zeros
```

```

theta_0 = np.zeros(trainX_with_bias.shape[1])

# Use scipy's minimize to find weights that minimize MAPE
res = minimize(mape, theta_0, args=(trainX_with_bias, trainY), method='BFGS')

# Optimal weights after training
theta_opt = res.x

theta_opt

```

```

Out[59]: array([-0.77801802,  1.10876973, -0.00160083, -0.04454795])

```

```

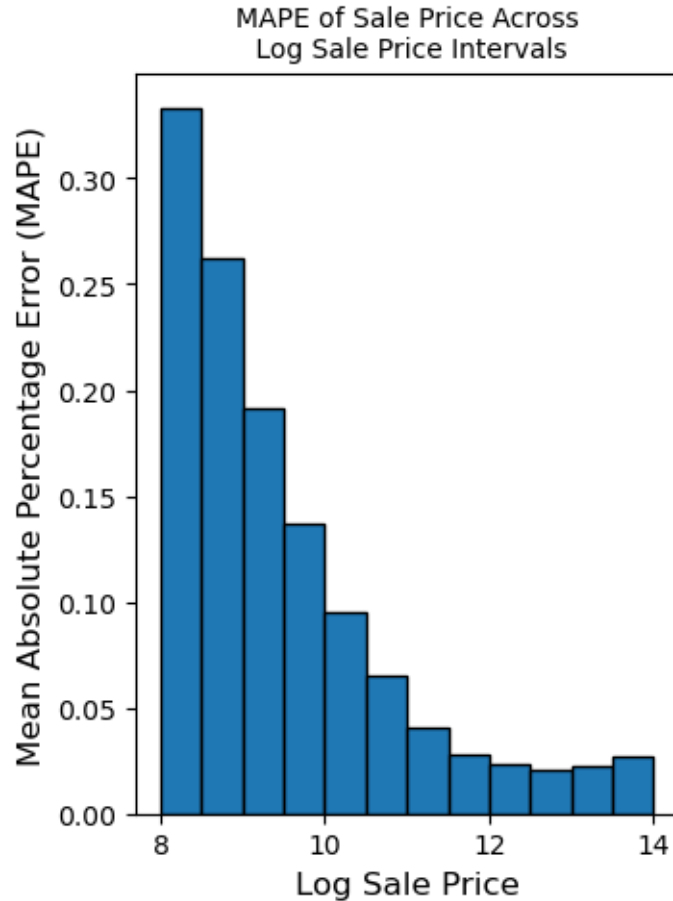
In [60]: new_preds_df = pd.DataFrame({
    'True Log Sale Price'      : trainY,
    'Predicted Log Sale Price': trainX_with_bias @ theta_opt,
    'True Sale Price'         : np.e ** trainY,
    'Predicted Sale Price'     : np.e ** (trainX_with_bias @ theta_opt)
})

plt.figure(figsize=(8, 5))
plt.subplot(1, 2, 1)

mape_values = []
for i in np.arange(8, 14, 0.5):
    mape_values.append(mape_interval(new_preds_df, i, i + 0.5))

plt.bar(x=np.arange(8.25, 14.25, 0.5), height=mape_values, edgecolor='black', width=0.5)
plt.title('MAPE of Sale Price Across\n Log Sale Price Intervals', fontsize=10)
plt.xlabel('Log Sale Price')
plt.ylabel('Mean Absolute Percentage Error (MAPE)')
plt.xticks(fontsize=10)
plt.yticks(fontsize=10);

```



What can you infer from this graph using our new custom error metric? Write your findings in the space below in three to five sentences. Consider addressing the following points:

- How does this metric differ from RMSE?
- What is the purpose of this custom error function?
- How does it relate to the idea of underpriced expensive housing mentioned in Question 6a?
- Why would this potentially be better than the RMSE in terms of the CCAO dataset?

MAPE looks at relative error $|y - \hat{y}| / y$ while RMSE looks at absolute squared error. That means RMSE is dominated by expensive homes (big dollar misses), but MAPE weights each home proportionally to its value. In our plot, MAPE is highest for low log-price bins, which lines up with the underpriced/overpriced pattern from Q6 – cheaper homes suffer larger percentage mistakes even if the dollar miss isn't huge. For the CCAO context, that's important as tax burden is felt more acutely for lower-priced homes, so a fairness metric that highlights who is hurt proportionally is more informative than RMSE alone. In short, MAPE helps surface regressive behavior that RMSE can hide.

